

$$= \frac{\quad}{32} \times 100 = 25$$

Though the standard deviations of both the groups are same, the coefficient of variation is higher for the group II. Hence the group II with mean age 10 shows more variability.

3.7. Lorenz Curve

It is a graphical method for measuring variability. It was devised by Lorenz for measuring the inequalities for the distribution of wealth. But it can be used with equal advantage to compare the distribution of profits among business and such other things.

Procedure for drawing Lorenz curve:

1. The size of the items ($x \text{ values} \times \text{frequency}$) and frequency are both cumulated. Percentages are obtained for these values.
2. On x - axis, start from 0 to 100 and take the percentages of cumulated frequencies.
3. On y - axis, take from 0 to 100 and consider the percentages of the cumulated values of the variable.
4. Draw the line joining O (0, 0) and P (100, 100). The line joining OP makes an angle 45° with x - axis. This line is known as the 'line of equal distribution'.

5. Plot percentages of cumulated size of the items against the corresponding percentages of cumulated frequencies.
6. Join the plotted points as a free hand curve. This curve will always lie on or below the line of equal distribution OP.

The greater the variability for the data, the greater is the distance of the curve from the line OP. The curve drawn is known as Lorenz curve for the data.

Problem 1: *The following data gives the income in rupees for the individuals in two cities. Draw the Lorenz curves and comment on the distribution of income in both the cities.*

Income :	0 – 1000	1000 – 5000	5000 – 20000	20000 – 50000	50000 – 1 Lakh
City A :	22	78	124	24	9
City B :	25	98	116	20	15

Solution

The calculations for drawing Lorenz curves for city A and city B are as follows:

City A:

city A. Hence the city B has more variability than city A.
distribution of wealth.

3.8. Box-Whisker Plot

A box plot or box-whisker plot is a set of five summary measures of the set of data; they are (i) median (ii) lower quartile (iii) upper quartile (iv) smallest observation (v) largest observation.

What makes the box plot (also called box-whisker-plot) a very useful graph is that it shows all of the above five numbers. Thus, it only takes some seconds to reveal the central tendencies and dispersion of one's data. Additionally, a box plot may also indicate any outliers.

For a given set of data, construction of box plot starts with the calculation of first, second and third quartiles Q_1 , Q_2 , and Q_3 of the data. Quartiles are the values which divide the entire data into four equal parts. Consider the data set contains N observations. To identify Q_1 , Q_2 , and Q_3 , first we arrange the data in ascending order of magnitude, then locate the value which is coming in the $\left(\frac{N}{4}\right)^{th}$,

$\left(\frac{N}{2}\right)^{th}$ and $\left(\frac{3N}{4}\right)^{th}$ position. These values are respectively Q_1 , Q_2 , and Q_3 .

Let us consider the following data showing the duration rounded in hours of 45 hospital patients slept following the administration of a certain anesthetic

6,10,12,4,8,7,6,8,5,12,11,3,8,7,2,13,8,4,4,5,5,8,7,7,3,2,3,8,9,6,7,10,3,
6,5,5,3,4,17,9,4,7,7,11 and 8.

Arranging the data in ascending order of magnitude, it is as follows: